3. Design Zero-shot, One-shot, and Few-shot prompts to summarize the following article into 50 words. Compare the generated summaries based on accuracy, completeness, and readability.

Aim

To implement Zero-shot, One-shot, and Few-shot prompting techniques using Python to summarize an article into 50 words and compare the generated summaries based on accuracy, completeness, and readability.

Algorithm

1. Start the program.
2. Import the required AI library.
3. Load a pre-trained text-generation model.
4. Enter the article to be summarized.
5. Design a Zero-shot prompt without examples.
6. Design a One-shot prompt with one example.
7. Design a Few-shot prompt with two examples.
8. Generate summaries using the AI model.
9. Compare the summaries based on accuracy, completeness, and readability.
10. Stop the program.

Code

```
from transformers import pipeline
ai = pipeline("text2text-generation",
        model="google/flan-t5-small")
article = """Artificial Intelligence is transforming
healthcare by helping doctors detect diseases,
analyze medical images, and predict patient risks.
AI supports drug discovery and personalized
treatment. Wearable devices monitor patient
health continuously. However, AI systems require
```

```
data privacy, security, and human supervision."""
zero = f"""Summarize this article in exactly
50 words. Include all important points:
{article}"""
one = f"""Example:
Text: AI helps students learn through
personalized education.
Summary: AI improves education by providing
personalized learning experiences.
Now summarize this article in 50 words:
{article}"""
few = f"""Example 1:
Text: AI improves agriculture through crop monitoring.
Summary: AI helps farmers monitor crops.
Example 2:
Text: AI improves transport through traffic prediction.
Summary: AI helps predict traffic conditions.
Now summarize this article in 50 words:
{article}"""
prompts = {
    "Zero-shot": zero,
    "One-shot": one,
    "Few-shot": few
}
for name, prompt in prompts.items():
    result = ai(prompt, max_new_tokens=100,
                do_sample=False)
```

```
    summary = result[0]["generated_text"]
    print("\n", name, "Summary:")
    print(summary)
    print("Word Count:", len(summary.split()))
```

Input

Article: Artificial Intelligence is transforming healthcare by helping doctors detect diseases, analyze medical images, and predict patient risks. AI supports drug discovery and personalized treatment. Wearable devices monitor patient health continuously. However, AI systems require data privacy, security, and human supervision.

Output

```
 Zero-shot Summary:
Summarize this article in exactly
50 words. Include all important points:
Artificial Intelligence is transforming
healthcare by helping doctors detect diseases,
analyze medical images, and predict patient risks.
AI supports drug discovery and personalized
treatment. Wearable devices monitor patient
health continuously. However, AI systems require
data privacy, security, and human supervision.
Word Count: 51

 One-shot Summary:
Example:
Text: AI helps students learn through
personalized education.
Summary: AI improves education by providing
personalized learning experiences.

Now summarize this article in 50 words:
Artificial Intelligence is transforming
healthcare by helping doctors detect diseases,
analyze medical images, and predict patient risks.
AI supports drug discovery and personalized
treatment. Wearable devices monitor patient
health continuously. However, AI systems require
data privacy, security, and human supervision.
Word Count: 65
```

```
 Few-shot Summary:
Example 1:
Text: AI improves agriculture through crop monitoring.
Summary: AI helps farmers monitor crops.

Example 2:
Text: AI improves transport through traffic prediction.
Summary: AI helps predict traffic conditions.

Now summarize this article in 50 words:
Artificial Intelligence is transforming
healthcare by helping doctors detect diseases,
analyze medical images, and predict patient risks.
AI supports drug discovery and personalized
treatment. Wearable devices monitor patient
health continuously. However, AI systems require
data privacy, security, and human supervision.
Word Count: 77
```

Result

Thus, Zero-shot, One-shot, and Few-shot prompting techniques were implemented for AI-based article summarization. The generated summaries can be compared based on accuracy, completeness, and readability.